Q20 Comprehension Check

Question: A colleague says: "LLMs are completely different from older neural networks — backprop doesn't really apply, they learn differently, and GPU optimizations for CNNs are useless for them." Identify each claim as correct or incorrect, with one sentence of justification each.

Answer: All three are incorrect. (1) LLMs are trained with backpropagation exactly like older networks — same reverse-mode autodiff, same optimizer family. (2) They learn by gradient descent on a loss like any deep model; the *loss* (next-token prediction) is different, not the learning rule. (3) At the hardware level both are dominated by dense matmuls; tensor cores and GEMM kernels tuned for convnets transfer directly to Transformer matmuls (and vice versa).